Voronoi Homogeneity Analysis of Categorical Data

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TBD

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1 Introduction

1.1 Homogeneity Analysis

Homogeneity Analysis (HGA), a.k.a. *Multiple Correspondence Analysis (MCA)*, can be introduced and implemented in many different ways[1]. We refer to Guttman (1941), Tenenhaus and Young (1985), Bekker and De Leeuw (1988), Gifi (1990), Greenacre and Blasius (2006), Le Roux and Rouanet (2010), Husson et al. (2016), and De Leeuw (2023) for a bouquet of discussions and interpretations of the equations and algorithms defining the HGA technique.

A *variable* is a function that maps on a (finite or infinite) domain of *objects* into a (finite or infinite) codomain of *values*. In HGA the data are the values of m variables on the same *carrier*, a finite subset the domain consisting of n objects. Each variable j has a finite number k_j of *categories*. The categories constitute the *range* of the variable for the n objects in the carrier. Note that all variables have the same carrier. Each object maps into a unique category, and each category contains a subset of objects from the carrier. We also say that a category has a number of objects that "belong to" the category.

Here is a small example with $n = 10$ and $m = 3$, taken from Gifi (1990), page 65. In the github voronoi repository these data are in data/small.R.

```
   V1 V2 V3
1   a  p  u
2   b  q  v
3   a  r  v
4   a  p  u
5   b  p  v
6   c  p  v
7   a  p  u
8   a  p  v
9   c  p  v
10  a  p  v
```

The key mathematical object in HGA is the *indicator matrix*, a classical way to code categorical data[2]. For a variable j with k_j categories it is an $n \times k_j$ binary matrix G_j in which each row i contains a single element equal to one, indicating into which category the object i maps for variable j. The cross product $G_j'G_j$ is diagonal, with the *marginal frequencies* of the categories on the diagonal. For two different variables the crossproduct $G_j'G_\ell$ is the $k_j \times k_\ell$ *cross table*

[1] Since PCA and MCA have three-letter acronyms I added the G of Gifi to the HA of Homogeneity Analysis.

[2] Since the indicator matrix is the core of the Gifi system we avoid the irreverent alternative "dummy coding".

or *contingency table* of the two variables. Here are the three indicator matrices of our small example, next to each other.

```
        a b c    p q r    u v
 [1,] || 1 0 0 || 1 0 0 || 1 0 ||
 [2,] || 0 1 0 || 0 1 0 || 0 1 ||
 [3,] || 1 0 0 || 0 0 1 || 0 1 ||
 [4,] || 1 0 0 || 1 0 0 || 1 0 ||
 [5,] || 0 1 0 || 1 0 0 || 0 1 ||
 [6,] || 0 0 1 || 1 0 0 || 0 1 ||
 [7,] || 1 0 0 || 1 0 0 || 1 0 ||
 [8,] || 1 0 0 || 1 0 0 || 0 1 ||
 [9,] || 0 0 1 || 1 0 0 || 0 1 ||
[10,] || 1 0 0 || 1 0 0 || 0 1 ||
```

We now deviate from the usual Gifi and French approaches (Husson et al. (2016)) and define HGA as a form of non-metric multidimensional scaling, using the *stress* loss function (Kruskal (1964a), Kruskal (1964b)). Define[3]

$$\sigma(X, Y_1, \cdots, Y_m, \Delta_1, \cdots, \Delta_m)) := \sum_{j=1}^{m}\sum_{i=1}^{n}\sum_{\ell=1}^{k_j} w_{i\ell}^j(\delta_{i\ell}^j - d(x_i, y_\ell^j))^2, \tag{1}$$

with $w_{i\ell}^j$ given non-negative weights, $d(x_i, y_\ell^j)$ the *Euclidean distance* between row i of X and row ℓ of Y_j, and $\delta_{i\ell}^j$ the corresponding *disparity*.

In classical HGA there are no weights, i.e. all $w_{i\ell}^j$ in (1) are equal to one. We have added weights to the definition of HGA so that, for example, we can easily incorporate missing data.

Disparities are required to satisfy the measurement restrictions implied by the data. In HGA this means that we require that $\delta_{i\ell}^j = 0$ whenever $g_{i\ell}^j = 1$, or, using the elementwise product $\circledast$ of matries, $\Delta_j \circledast G_j = 0$.

HGA minimizes (1) over the $n \times p$ configurations X satisfying $X'X = I$, over the Y_j, and over the Δ_j that satisfy the measurement restrictions.

Define

$$\sigma_\star(X, Y_1, \cdots, Y_m) := \min_{\Delta_1, \cdots, \Delta_m} \sigma(X, Y_1, \cdots, Y_m, \Delta_1, \cdots, \Delta_m). \tag{2}$$

[3]The symbol $:=$ is used for definitions

The minimizing disparity $\delta_{i\ell}^j$ is equal to zero if $g_{i\ell}^j = 1$ and equal to $d(x_i, y_\ell^j)$ otherwise. Consequently

$$\sigma_\star(X, Y_1, \cdots, Y_m) = \sum_{j=1}^m \sum_{i=1}^n \sum_{\ell=1}^{k_j} w_{i\ell}^j g_{i\ell}^j d^2(x_i, y_\ell^j). \tag{3}$$

Note that the value of (3) does not depend on the $w_{i\ell}^j$ for which $g_{i\ell}^j = 0$. This implies that we can actually forget about the subscript ℓ in $w_{i\ell}^j$ and simply use w_i^j.

We can make a graphical representation of (3). For variable j we plot the n points x_i and the k_j points y_ℓ^j in a joint plot (or biplot). We then connect, with straight lines, each category point with the object points that belong to it. For each category this defines a figure in the plot which we call a *star*. The *size* of a star is the weighted sum of squares of the length of its lines. The loss function (3) for variable j is the weighted sum of squares of the length of all n lines in this plot, or equivalently the sum of the sizes of the k_j stars. The total loss is the sum of star-sizes over all m plots. Thus HGA aims to place the object-points and category-points in space in such a way that the stars are as small as possible, i.e. objects are as close as possible to the categories they belong to.

In matrix notation

$$\sigma_\star(X, Y_1, \cdots, Y_m) = \sum_{j=1}^m \text{tr}\, \{X'W_jX - 2\, X'W_jG_jY_j + Y_j'D_jY_j\} \tag{4}$$

Here W_j is a diagonal matrix with the w_i^j and D_j is the diagonal matrix of column-sums of W_jG_j. Note that in classical HGA, with $W_j = I$ for all j, we have $D_j = G_j'G_j$.

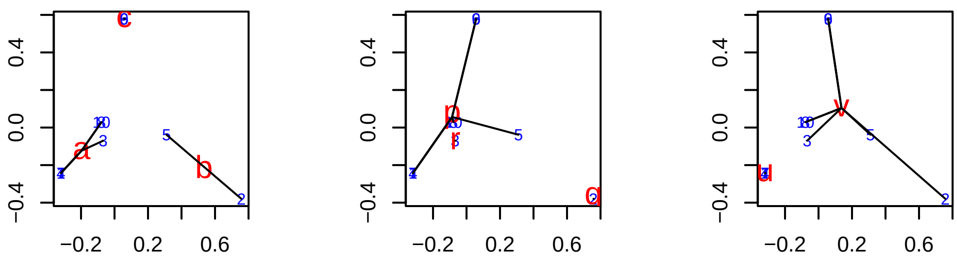
Next define

$$\sigma_{\star\star}(X) := \min_{Y_1, \cdots, Y_m} \sigma_\star(X; Y_1, \cdots, Y_m). \tag{5}$$

The minimum of (4) over the Y_j is attained at $\hat{Y}_j := D_j^{-1}G_j'W_jX$. Category-points are the weighted centroids of the object-points belonging to the category. This makes stars actually look like stars. If we substitute these optimal $\widehat{Y}_j$ in (3) we see that $\sigma_{\star\star}(X)$ is indeed equal to the weighted sum of squares of all lines defining the stars. Minimizing $\sigma_{\star\star}(X)$ over $X'X = I$, and thus minimizing (1) over (X, Y_j, Δ_j), defines HGA.

Computing the solution turns out to be a straightforward eigenvalue-eigenvector problem. The details are in the publications we referenced at the beginning of this section.

A HGA in two dimensions produces of our small example the following three star plots. Objects are numbered and in BLUE, categories have bigger labels and are in RED. Note that there are ten objects, but the sets of objects $(1, 4, 7)$, $(8, 10)$, and $(6, 9)$ have the same *profile* and consequently get mapped into the same object-point. Thus there are at most six different object-points. The analysis minimizes the sum of squares of the length of all black lines in the three plots, under the condition that the matrix of object scores is column-centered and orthonormal. We can think of the results as a form of *graph drawing*, where the graph is the union of the m bipartite graphs of all m "belonging to" relations (De Leeuw and Michailidis (2000), Michailidis and De Leeuw (2001)).



1.2 Voronoi Homogeneity Analysis

Although using loss function (1) makes HGA a form of non-metric multidimensional scaling there are some important differences with the more familiar MDS versions minimizing stress. There is no local minima problem in HGA, because it turns out to be a symmetric eigenvalue problem. In HGA, unlike in MDS, we impose the constraint $X'X = I$. Also, and perhaps most importantly, in HGA we do not really expect the minimum loss function value to be zero, or even close to zero. The loss function measures deviations from a very unrealistic "model". Zero loss would imply all object-points coincide with all category-points they belong to. This can only be realized, except in trivial cases, if all object and category-points map into a single point, which is prohibited by the orthonormality constraint on the object scores. Thus we get a perfectly determinate and stable solution, at low computational cost, but by imposing strong constraints on the disparities and a rather arbitrary normalization. We can get around this seemingly unfortunate situation by admitting we are not fitting or testing a "model", but we are creating "best" drawings of the bipartite graphs that constitute the data.

The lack of a realistic model makes it interesting to continue use the more general loss function

$$\sigma(X, Y_1, \cdots, Y_m, \Delta_1, \cdots, \Delta_m)) := \sum_{j=1}^{m} \sum_{i=1}^{n} \sum_{\ell=1}^{k_j} w_{i\ell}^j (\delta_{i\ell}^j - d(x_i, y_\ell^j))^2, \tag{6}$$

with the constraints that if $g_{i\ell}^j = 1$ then merely $\delta_{i\ell}^j \leq \delta_{i\nu}^j$ for all $\nu = 1, \cdots, k_j$. This is significantly weaker than the HGA constraints, which require the mn disparities $\delta_{i\ell}^j$ to be zero. If object i belongs to category ℓ of variable j we do no longer require the object-point to coincide with the category-point, but we merely require that the object-point is at least as close to category-point ℓ as to any of the other $k_j - 1$ category-points of that variable.

Voronoi Homogeneity Analysis (VHA) of categorical data minimizes loss function (1) over X, over the Y_j and over the Δ_j satisfying the relaxed ordinal constraints. It is consequently another form of non-metric (ordinal) multidimensional scaling. To be more precise, we have m linked non-metric multidimensional unfolding problems for the binary matrices G_j, which are "linked" because they share the same X.

Because of the different ordinal constraints on the disparities, the plots corresponding with this analysis are different. As in HGA there is a plot for each variable j with both n object-points and k_j category-points. The k_j category-points are the *generators* of k_j *Voronoi regions* that partition the space. Voronoi region $\mathcal{V}_\ell^j$ is the region of space where the points are at least as close to generator y_ℓ^j as to any of the other generators of variable j. And loss (4) is zero (for a variable) if all object-points are in the "correct" Voronoi region, i.e. in the Voronoi region of the generator corresponding with the category they belong to.

One (inefficient, but conceptually easy) way of constructing Voronoi regions in the plane is to draw the perpendicular bisectors of the (hypothetical) lines connecting all pairs of category-points of a variable. In higher dimensions $p > 2$ the bisectors are planes of dimension $p - 1$. The convex polyhedral regions thus formed, some bounded and some unbounded, define the Voronoi regions[4]. The literature on properties and computation of Voronoi regions is large, and we refer to the textbooks of Okabe et al. (2000) and Aurenhammer et al. (2013).

Since the category-point itself is obviously in its own Voronoi region and since Voronoi regions are convex it follows that if we make stars by connecting category-points with object points as in HGA then we actually require each star to be in its Voronoi region. It is no longer true that a category-point is the centroid of points of the objects belonging to the category. The category-point can even be outside the convex hull of the object-points belonging to the category.

The constraint $X'X = I$ from HGA is no longer imposed directly, but we somehow have to deal with the fact that in VHA the minimum of loss function (1) is always zero. The trivial solution, in which all points and regions collapse into a single point and all Δ_j are zero, is always available and not excluded by the constraints. But, perhaps more urgently, we have to deal with the types of degenerate solutions familiar from multidimensional unfolding (Heiser (1981), De Leeuw (1983), Busing (2010)). For example, we can collapse all category-points of a variable into a single point and put all object-points on a sphere with the category-point at its center. Then all distances between category and object-points are the same and we can choose all disparities equal to that distance value and obtain loss zero.

We will use some constraints that tend to counteract degeneracy. Also, we always start our iterations with the HGA analysis, which in most cases already has small loss. Then we perform iterations until either the distances do not change from one iteration to the next (up to some

[4] In the plane a Voronoi region is unbounded if and only if its generator is on the boundary of the convex hull of all generators.

epsilon) or the loss becomes zero (up to some epsilon) or we have reached the maximum number of iterations allowed.

2 Algorithm

We use an *Alternating Least Squares* or *ALS* algorithm, similar to the approach used in many other non-metric scaling methods, to minimize loss (5). For context, see Borg and Groenen (2005). We alternate minimization over $\Delta_1, \cdots, \Delta_m$ for given X and $Y_1 \cdots Y_m$ with minimization over X and $Y_1 \cdots Y_m$ for given $\Delta_1, \cdots, \Delta_m$. Using superscript (μ) for the iteration index we can write for an ALS iteration

$$(\Delta_1^{(\mu)}, \cdots, \Delta_m^{(\mu)}) = \underset{\Delta_1, \cdots, \Delta_m}{\text{argmin}}\ \sigma(X^{(\mu)}, Y_1^{(\mu)}, \cdots, Y_m^{(\mu)}, \Delta_1, \cdots, \Delta_m), \tag{7a}$$

$$(X^{(\mu+1)}, Y_1^{(\mu+1)}, \cdots, Y_m^{(\mu+1)}) = \underset{X, Y_1, \cdots, Y_m}{\text{argmin}}\ \sigma(X, Y_1, \cdots, Y_m, \Delta_1^{(\mu)}, \cdots, \Delta_m^{(\mu)}). \tag{7b}$$

The first substep (7a) of iteration (μ) solves a monotone regression problem for a simple partial order, in which one of k_j elements must be smaller than the other $k_j - 1$ elements. The second substep (7b) solves a metric multidimensional scaling problem. This second subproblem cannot be solved completely, because it would need an infinite iterative process by itself. Thus the second substep (7b) consists of a finite number of *inner iterations* within the second substep of a single ALS *outer iteration*.

2.1 Initial Configuration

In all our analysis we use HGA as the initial configuration for both object-scores and category-scores. The algorithm for HGA is the ALS method used in Gifi (1990) to compute the truncated SVD. The repository includes mca.R

2.2 Monotone Regression

The nm monotone regressions are applied for each object i and each variable j. Forgetting about all these indices for the moment we need to minimize the sum of squares $\|\delta - d\|^2$ over δ, where one of the elements, say δ_ν, must be smaller than or equal to all others.

This can be solved by using monotone regression with the primary approach to ties (Kruskal (1964b), De Leeuw (1977)), treating all δ_ℓ with $\ell \neq \nu$ as ties.

2.3 Majorization

We study the off-diagonal smacof algorithm for MDS, without taking into account that the loss function (1) comes from n observations on m categorical variables. This improves and generalizes the smacof treatment of the metric unfolding problem in Heiser and De Leeuw (1979).

We rewrite stress as

$$\sigma(X,Y):=\sum_{i=1}^n\sum_{k=1}^K w_{ik}(\delta_{ik}-d(x_i,y_k))^2. \tag{8}$$

The distance is

$$d(x_i,y_k)=\sqrt{(X'e_i-Y'e_k)'(X'e_i-Y'e_k)}. \tag{9}$$

We will construct a majorization of (8) at $\tilde X$ and $\tilde Y$. Think of $\tilde X$ and $\tilde Y$ as the current best estimates in a sequence of iterations. By Cauchy-Schwartz,

$$d(x_i,y_k)\geq\frac{1}{d(\tilde x_i,\tilde y_k)}(X'e_i-Y'e_k)'(\tilde X'e_i-\tilde Y'e_k). \tag{10}$$

Define the $n\times K$ matrix-valued function $B(X,Y)$ by

$$b_{ik}(X,Y):=\begin{cases}w_{ik}\frac{\delta_{ik}}{d(x_i,y_k)}&\text{if } d(x_i,y_k)>0,\\ 0&\text{if } d(x_i,y_k)=0.\end{cases} \tag{11}$$

Also define the diagonal matrices $B_\star(X,Y)$ with the row-sums of $B(X,Y)$ and $B^\star(X,Y)$ with the column-sums.

If we define, as we usually do in smacof,

$$\rho(X,Y):=\sum_{i=1}^n\sum_{k=1}^K w_{ij}\delta_{ij}d_{ij}(x_i,y_k) \tag{12a}$$

hen $\rho(X,Y)\geq\rho(X,Y;\tilde X,\tilde Y)$, where

$$\rho(X,Y;\tilde X,\tilde Y):=\text{tr}\,X'B_\star(\tilde X,\tilde Y)\tilde X-\text{tr}\,X'B(\tilde X,\tilde Y)\tilde Y-\text{tr}\,Y'B'(\tilde X,\tilde Y)\tilde X+\text{tr}\,Y'B^\star(\tilde X,\tilde Y)\tilde Y. \tag{12b}$$

Also define

$$\eta^2(X,Y):=\sum_{i=1}^n\sum_{k=1}^K w_{ik}\delta_{ij}d_{ij}^2(x_i,y_k)=\text{tr}\,X'W_\star X-2\,\text{tr}\,X'WY+\text{tr}\,Y'W^\star Y, \tag{12c}$$

with $W_\star$ and $W^\star$ the diagonal matrices with row-sums and column-sums of W, as well as the constant

$$C := \sum_{i=1}^n \sum_{k=1}^K w_{ik}\delta_{ik}^2 \tag{12d}$$

With these definitions

$$\sigma(X,Y) = C - 2\,\rho(X,Y) + \eta^2(X,Y), \tag{13}$$

and $\sigma(X,Y) \leq \sigma(X,Y;\tilde{X},\tilde{Y})$, where

$$\sigma(X,Y;\tilde{X},\tilde{Y}) := C - 2\,\rho(X,Y;\tilde{X},\tilde{Y}) + \eta^2(X,Y), \tag{14}$$

with equality if $X = \tilde{X}$ and $Y = \tilde{Y}$. In a majorization (or MM) iteration we minimize $\sigma(X,Y;\tilde{X},\tilde{Y})$ over X and Y. If the minimum is attained at $\tilde{X}$ and $\tilde{Y}$ we stop the algorithm, because we have found a stationary point. If not we have found updates X^+ and Y^+ such that

$$\sigma(X^+,Y^+) \leq \sigma(X^+,Y^+;\tilde{X},\tilde{Y}) < \sigma(\tilde{X},\tilde{Y};\tilde{X},\tilde{Y}) = \sigma(\tilde{X},\tilde{Y}). \tag{15}$$

The stationary equations for minimizing $\sigma(X,Y;\tilde{X},\tilde{Y})$ are

$$X = W_\star^{-1}(WY + B_\star(\tilde{X},\tilde{Y})\tilde{X} - B(\tilde{X},\tilde{Y})\tilde{Y}), \tag{16a}$$

$$Y = \{W^\star\}^{-1}(W'X + B^\star(\tilde{X},\tilde{Y})\tilde{Y} - B'(\tilde{X},\tilde{Y})\tilde{X}). \tag{16b}$$

Alternatively

$$\begin{bmatrix} W_\star & -W \\ -W' & W^\star \end{bmatrix}\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} B_\star(\tilde{X},\tilde{Y}) & -B(\tilde{X},\tilde{Y}) \\ -B'(\tilde{X},\tilde{Y}) & B^\star(\tilde{X},\tilde{Y}) \end{bmatrix}\begin{bmatrix} \tilde{X} \\ \tilde{Y} \end{bmatrix} \tag{17}$$

The next step in each iteration could be to directly solve the system (17) for X and Y. This approach has the disadvantage that the order is $n+m$, which can be quite large in applications. We also have to take into account that the coefficient matrix of (17) is singular. Alternatively we can substitute the expression for X from (16a) in (16b) and then solve the resulting equation for Y. Or we can solve the system in (16a) and (16b) iteratively by ALS. Choose a Y, then compute X from (16a), then use that X in (16b) to find a new Y, and so on.

In this paper we use an approach similar to ALS, which could be called MALS for Majorization by Alternating Least Squares. In the first step of a MALS cycle we majorize $\sigma(X,\tilde{Y})$ and minimize $\sigma(X,\tilde{Y};\tilde{X},\tilde{Y})$. Thus we treat the loss as a function of X only, and by the earlier results in this section we have

$$X^+ = W_\star^{-1}(W\tilde{Y} + B_\star(\tilde{X},\tilde{Y})\tilde{X} - B(\tilde{X},\tilde{Y})\tilde{Y}). \tag{18a}$$

The next step, in the same MALS iteration, is to us this new X^+ and minimize $\sigma(X^+, Y; X^+, \tilde{Y})$ over Y, giving

$$Y^+ = \{W^\star\}^{-1}(W'X^+) + B^\star(X^+, \tilde{Y})\tilde{Y} - B'(X^+, \tilde{Y})X^+) \tag{18b}$$

Then proceed to the next two-step MALS iteration, using the new X^+ and Y^+.

One advantage of MALS, shared with ALS, is that the only matrix inversion needed is of thee diagonal weight marginals. Also there is no need to work with square symmetric matrices of order $n + m$. But the main advantage of MALS is that it can easioly incorporate various constraints on X and Y. Remember that

$$\sigma(X, \tilde{Y}; \tilde{X}, \tilde{Y}) = \text{tr } X'W_\star X - 2 \text{ tr } X'\{W\tilde{Y} - \text{tr } X'B_\star(\tilde{X}, \tilde{Y})\tilde{X} + \text{tr } X'B(\tilde{X}, \tilde{Y})\tilde{Y}\} + C, \tag{19}$$

where C is now used generically for terms not involving X. Define

$$\overline{X} := W\tilde{Y} - \text{tr } X'B_\star(\tilde{X}, \tilde{Y})\tilde{X} + \text{tr } X'B(\tilde{X}, \tilde{Y})\tilde{Y} \tag{20}$$

Then, after completing the square,

$$\sigma(X, \tilde{Y}; \tilde{X}, \tilde{Y}) = \text{tr } (X - \overline{X})'W_\star(X - \overline{X}) + C. \tag{21a}$$

Thus minimizing $\sigma(X, \tilde{Y}; \tilde{X}, \tilde{Y})$ over X that must satisfy some constraints means projecting $\overline{X}$ in the diagonal $W_\star$ metric on the set defined by the constraints. Of course in the same way

$$\sigma(\tilde{X}, Y; \tilde{X}, \tilde{Y}) = \text{tr } (Y - \overline{Y})'W^\star(Y - \overline{Y}) + C. \tag{21b}$$

3 Constraints

3.1 Regression Constraints

In this paper we (optionally) use the constraints

$$\sum_{\ell=1}^{k_j} w_{i\ell}^j(\delta_{i\ell}^j - \overline{\delta}_{ij})^2 = w_{i\star}^j, \tag{22}$$

for all i and j, with $\overline{\delta}_{ij}$ the weighted mean of the $\delta_{i\ell}^j$. In addition, of course, the $\delta_{i\ell}^j$ must satisfy the ordinal constraints. Constraint (22) explicitly excludes the possibility that the $\delta_{i\ell}^j$ for given i and j are equal for all ℓ.

Suppose y is a vector with n elements. The problem we study is to minimize $\sigma(x) := \|x-y\|_W^2$ over $x \in K$ and $\|x - \overline{x}e\|_W^2$ is equal to some constant c. Here

- K is the cone of vectors defined by a number of inequalities of the form $x_i \leq x_j$,
- e is the vector with all n elements equal to one,
- W is a diagonal matrix of weights,
- $\|x\|_W^2 = x'Wx$ is the square of the W-norm of x,
- $\overline{x}$ is the weighted mean $e'Wx/e'We$.

Define $J := I - ee'W/(e'We)$ and $\tilde{x} = Jx$. Then $Jx = x - \overline{x}e$. Then $x'Jx = \tilde{x}'W\tilde{x}$ and $x \in K$ if and only if $\tilde{x} \in K$. Also $\tilde{x} = Jx$ if and only if $x = \tilde{x} + \beta e$. So the transformed problem is to minimize $\|\tilde{x} + \beta e - y\|_W$ over β and $\tilde{x} \in K$, with $e'W\tilde{x} = 0$ and $\tilde{x}'W\tilde{x} = c$. The minimizer for β is $\overline{y} := e'Wy/e'We$, and thus the equivalent "reduced" problem becomes minimizing $\|\tilde{x} - Jy\|_W^2$ over $\tilde{x} \in K$, $\tilde{x}'W\tilde{x} = c$, and $e'W\tilde{x} = 0$.

Forget about the constraint $e'W\tilde{x} = 0$ for now. If M is the monotone regression operator it follows that the solution for $\tilde{x}$ is $\sqrt{c}M(Jy)/\|M(Jy)\|_W$, which automatically satisfies $e'W\tilde{x} = 0$ (Bauschke et al. (2018)). Thus the solution of our original problem is

$$\hat{x} = \sqrt{c}\frac{M(y-\overline{y})}{\|M(y-\overline{y})\|_W} + \overline{y}e.$$

This can be rewritten in a perhaps more revealing way. Now $M(y - \overline{y}e) = M(y) - \overline{y}e$ and $e'WM(y) = e'Wy = e'We\overline{y}$. Thus $\|M(y-\overline{y})\|_W^2 = \|M(y)\|_W^2 - e'We\overline{M(y)}^2$ so that

$$\hat{x} = \frac{M(y) - \overline{y}e}{\|M(y-\overline{y})\|_W} + \overline{y}e = \frac{1}{\|M(y-\overline{y})\|_W}M(y) + \left\{1 - \frac{1}{\|M(y-\overline{y})\|_W}\right\}\overline{y}e$$

3.2 Configuration Constraints

Separation

3.2.1 Column-Centering

If we require $X'W_re = 0$ then the stationary equations are $W_r(X - \overline{X}) = W_re\lambda'$ and thus $X = \overline{X} + e\lambda'$ with $\lambda' = -e'W_r\overline{X}/e'W_re$ or

$$X = (I - \frac{ee'W_r}{e'W_re})\overline{X}$$

This is different from updating X as $J_n\tilde{X}$ and Y as $\tilde{Y} - n^{-1}ee'\tilde{X}$. With this alternative update, in which we subtract the column-means of $\tilde{X}$ from both $\tilde{X}$ and $\tilde{Y}$, the distances are the same before and after centering. Thus the distances in all iterations, and consequently the loss function values, are the same with or without this type of centering.

3.2.2 Orthogonality

If we require $X'W_\star X = I$ we have the stationary equations $\overline{X} = XM$, with M a symmetric matrix of Lagrange multipliers. Thus

$$X = \text{procrustus}(\overline{X}) := \overline{X}(\overline{X}'W_\star\overline{X})^{-\frac{1}{2}}.$$

If we require both $X'W_\star X = I$ and $X'W_\star e = 0$ we have

$$X = \text{procrustus}((I - \frac{ee'W_\star}{e'W_\star e})\overline{X})$$

3.2.3 Rank Constraints

One of the main themes in Gifi (1990) is that the $k_j \times p$ matrices of category-scores can be constrained to be of rank $r < p$. We will restrict ourselves in this paper to $r = 1$, which defines what Gifi calls *single quantifications*. If all variables are restricted to have single quantifications then a HGA becomes equivalent to a PCA of the transformed variables.

In VHA the Voronoi regions of single variables become parallel strips. The category-points of a variable are on a line through the origin, and the Voronoi lines or planes are parallel and perpendicular to the line with the category-points.

$$\text{tr }(Y-za')'W_c(Y-za')=-2z'W_cYa+z'W_cz.a'a$$

So rank one approximation of $\tilde{Y}+n^{-1}ee'(X-\tilde{X})$. If X is column-centered then of course $\tilde{Y}-ee'\tilde{X})$.

3.2.4 Centroid

A more complicated restriction is $Y=D^{-1}G'X$, possibly together with $X'X=I$ and $e'X=0$. Minimization over X and the Y_j is no longer separated. In fact, the *centroid constraints* effectively eliminate the Y_j from the optimization problem. We have to adapt the majorization method to take these new constraints into account.

If $Y_j=D_j^{-1}G_j'X$ then $d_{i\ll}^j(X,Y_j)$ only depends on X and we have

$$\{d_{i\ell}^j(X)\}^2=\text{tr }X'\left\{e_ie_i'-2\,\{d_\ell^j\}^{-1}g_\ell^je_i'+\{d_\ell^j\}^{-2}g_\ell^j\{g_\ell^j\}'\right\}X. \tag{23a}$$

Moreover for each pair X and $\tilde{X}$ we have, by Cauchy-=Schwartz,

$$d_{i\ell}^j(X)\geq\frac{1}{d_{i\ell}^j(\tilde{X})}\text{tr }X'\{e_ie_i'-2\,\{d_\ell^j\}^{-1}g_\ell^je_i'+\{d_\ell^j\}^{-2}g_\ell^j\{g_\ell^j\}'\}\tilde{X} \tag{23b}$$

Suppose R_W^j and C_W^j are diagonal matrices with row-sums and column=sums of W_j. Then

$$\sum_{i=1}^n\sum_{\ell=1}^{k_j}w_{i\ell}^j\{d_{i\ell}^j(X)\}^2=\text{tr }X'\left\{R_W^j-G_jD_j^{-1}W_j'-W_jD_j^{-1}G_j'+G_j(C_W^jD_j^{-2})G_j'\right\}X, \tag{24a}$$

and

$$\sum_{i=1}^n\sum_{\ell=1}^{k_j}w_{i\ell}^jd_{i\ell}^j(X)\geq\text{tr }X'\left\{R_{\tilde{B}}^j-G_jD_j^{-1}\tilde{B}_j'-\tilde{B}_j'D_j^{-1}G_j'+G_j(C_{\tilde{B}}^jD_j^{-2})G_j'\right\}\tilde{X}. \tag{24b}$$

In (24b) the matrix $\tilde{B}_j$ is defined as

$$\tilde{b}_{i\ell}^j:=\begin{cases}w_{i\ell}^j\frac{\delta_{i\ell}^j}{d_{i\ell}^j(\tilde{X})} & \text{if } d_{i\ell}^j(\tilde{X})>0,\\ 0 & \text{if } d_{i\ell}^j(\tilde{X})=0,\end{cases} \tag{25}$$

and $R_{\tilde{B}}^j$ and $C_{\tilde{B}}^j$ are the diagonal matrices of its row-sums and column-sums.

4 Software

4.1 Main

The main program voronoiHomogeneityAnalysis(), aliased as vha(), has the following arguments (with default values).

```
voronoiHomogeneityAnalysis <- function(theData,  # n x m matrix with categorical varia
                                       ndim = 2, # dimensionality
                                       wght = NULL,
                                       dnorm = FALSE,
                                       xcent = FALSE,
                                       xnorm = FALSE,
                                       yrank = ndim,
                                       itmax = 1000,
                                       eps = 1e-6,
                                       aps = 1e-6,
                                       verbose = TRUE)
```

- *theData* is an $n \times m$ matrix with n observations on $m$categorical variables.
- *ndim* is the number of dimensions of the solution.
- *wght* is either NULL or a list of $n \times k_j$ matrices. If wght is NULL all weights are one.
- *dnorm* if TRUE the monotone regression is normalized.
- *xcent* if TRUE object scores are W-centered.
- *xnorm* if TRUE object scores are W-orthonormal.
- *yrank* rank restrictions on category-scores. If the length of yrank is one it is expanded to a vector of length m.
- *itmax* is the maximum number of iterations.
- *eps* is the cutoff for the change in distances from one iteration to another.
- *aps* is the cutoff for the loss function value. If loss is smaller than *aps* the program stops.
- *verbose* if TRUE iteration information is printed to stdout.

The value returned by the program is a list.

```
  return(
    list(
      xmat = xnew,
      ymat = ynew,
      dmat = dnew,
      dhat = dhat,
```

```
      xini = xini,
      yini = yini,
      itel = itel,
      loss = snew,
      indi = indi,
      ncat = ncat,
      wght = wght
    )
  )
```

- *xmat* object scores, an $n \times p$ matrix, with p equal to *ndim*.
- *ymat* category-scores, a list of *m* matrices of dimension $k_j \times p$.
- *dmat* distances, a list of *m* matrices of dimension $n \times k_j$.
- *dhat* disparities, a list of *m* matrices of dimension $n \times k_j$.
- *xini* the HGA solution for the object scores,
- *yini* the HGA solution for the category-scores,
- *itel* number of iterations.
- *loss* final value of stress.
- *indi* indicators, a list of *m* matrices of dimension $n \times k_j$.
- *ncat* number of categories of the m variables.
- *wght* weights, a list of *m* matrices of dimension $n \times k_j$.

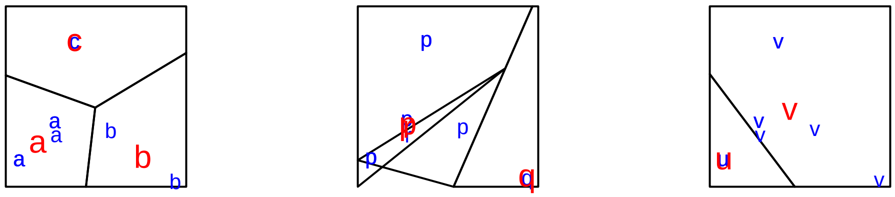
4.2 Subroutines

4.3 Plots

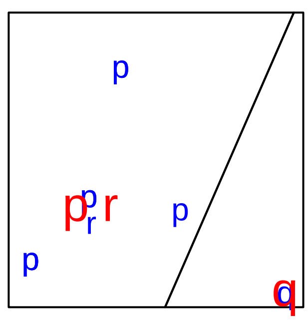
5 Examples

5.1 Small Example

We start with the small example we used before to illustrate HGA.



Loss is 0.0000000000 after 443 iterations. If we compare the plots from the Voronoi analysis with the star plots from the initial HGA for this example we see that the algorithm did not have to change much to obtain perfect fit. One thing that does change is that the iterations moves the category-points "p" and "r" for variable 2 very close to each other. At the point where we stop the iterations they are still distinguishable, and thus their perpendicular bisector is still a line in the Voronoi plot. If we would continue iterating "p" and "r" would coincide and there would only be two Voronoi regions: one containing category-point "q" and object-point 2, and one containing all other points. one containing all other.



5.2 GALO

Our next example use the GALO data from Peschar (1975). In the github voronoi repository the data are in data/galo.R. Subjects are 1290 children who in 1959/1960 were in sixth grade of primary school in the city of Groningen. Variables are SEX, IQ (in nine ordered categories), ADV (sixth grade teachers advice on the form of secondary education appropriate for the student), and SES. The categories of ADV are as follows (alphabetically).

- *Agr* Agricultural.
- *Ext* Extended primary (like middle school).
- *Gen* General (like high school).
- *Grls* Extended primary for girls (like middle school).
- *Man* Vocational, including housekeeping.
- *None* No secondary education.

The categories of SES are as follows (again alphabetically).

- *LowWC* Lower white collar.
- *MidWC* Middle white collar.
- *Prof* Professional.
- *Shop* Shopkeepers.
- *Skil* Skilled manual.
- *Unsk* Unskilled Manual.

We start with an HGA of the galo data. The initial value of the loss function is 0.3158339145. The loss function value is computed by using

$$\sigma_\star(X,Y_1,\cdots,Y_m):=\min_{\Delta_1,\cdots,\Delta_m}\sigma(X,Y_1,\cdots,Y_m,\Delta_1,\cdots,\Delta_m), \tag{26}$$

where X and the Y_j are the HGA solution, and where the minimum over the Δ_j is computed by requiring that $\delta^j_{i\nu}\leq\delta^j_{i\ell}$ if $g^j_{i\nu}=1$. This is *not* the HGA loss function (6), in which the minimum over the Δ_j is computed by requiring that $\delta^j_{i\ell}=0$ whenever $g^j_{i\ell}=1$. Thus we evaluate the HGA solution using the VGA loss function.

For any HGA or VGA solution we can compute a Shepard diagram by plotting the distances and disparities, pooled over all objects (individuals) and variables. The weak regression requirements imply that many disparities will be equal to the corresponding distances. This is illustrated by the diagonal of Figure 1.

The HGA solution can also be used to produce Voronoi plots, which are given in Figure 2 for IQ. We see the familiar horseshoe shape of the object score cloud (the Guttman effect, see De Leeuw (2023)). The category-scores are largely in order along the horseshoe, and as a consequence the Voronoi regions are unbounded and emanate from a point inside the horseshoe. There are some relatively monor deviations from that pattern. Specifically category 1 does not fit the pattern, but that is not too serious because there are only five individuals in that category. There are quite a few misclassifications, i.e. objects in the “wrong” Voronoi region.

The VHA solution has loss equal to 0.0000818320 after 1000 iterations. The voronoi function stopped because the maximum number of iterations was reached. The $d-\hat{d}$ plot in Figure 3 shows a near-perfect fit.

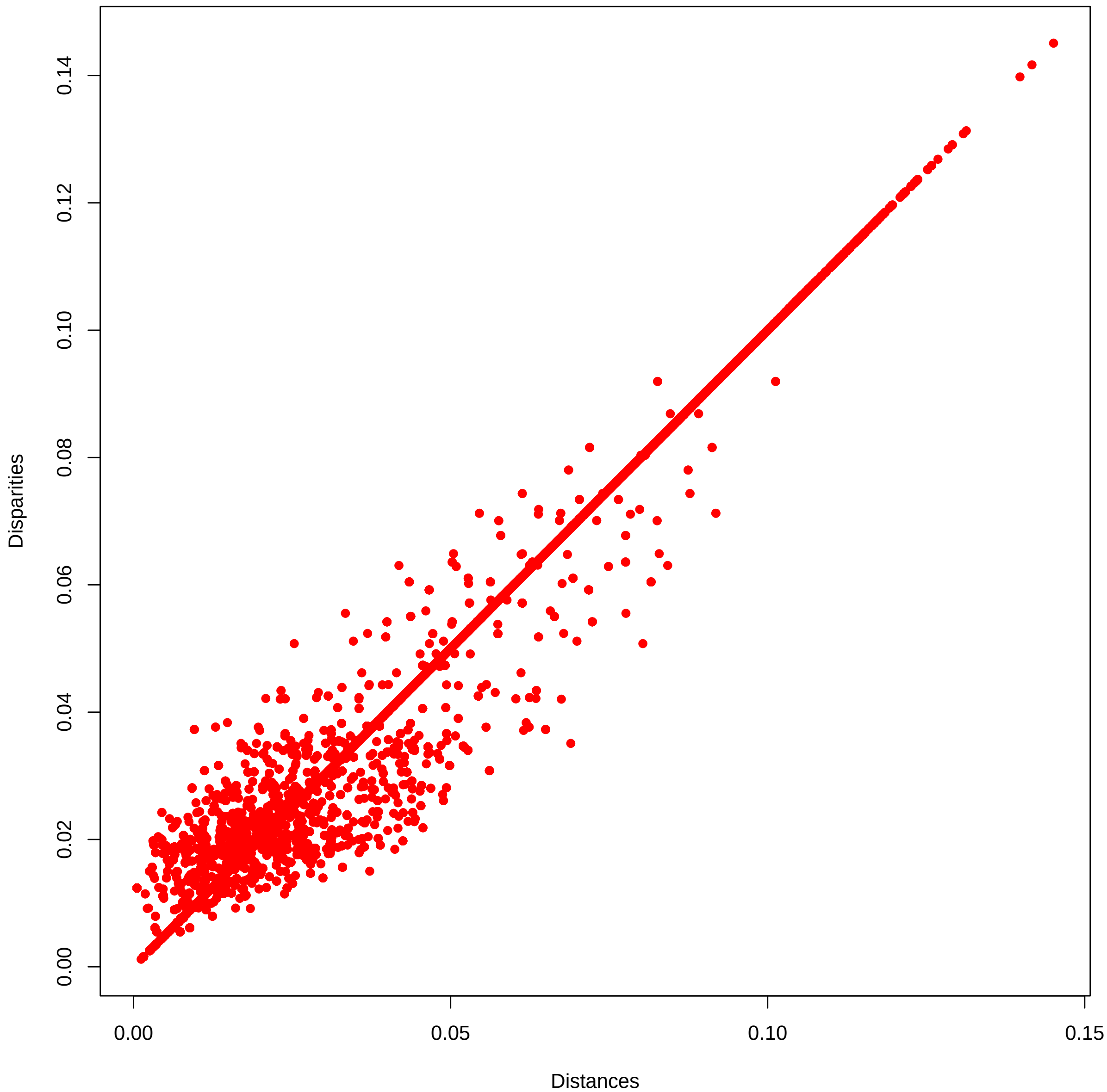


Figure 1: Galo D-Dhat Plot HGA

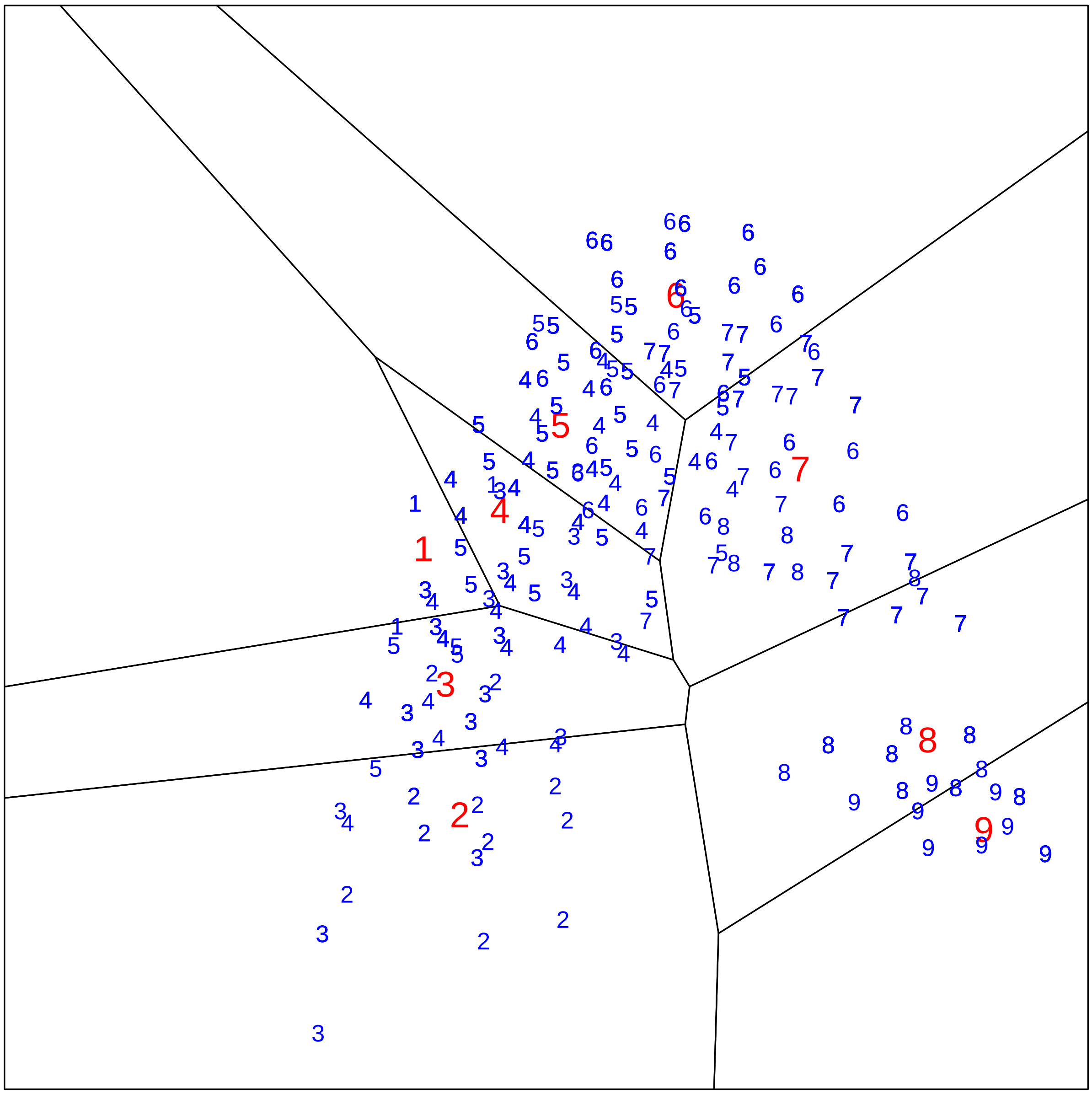


Figure 2: Galo IQ HGA

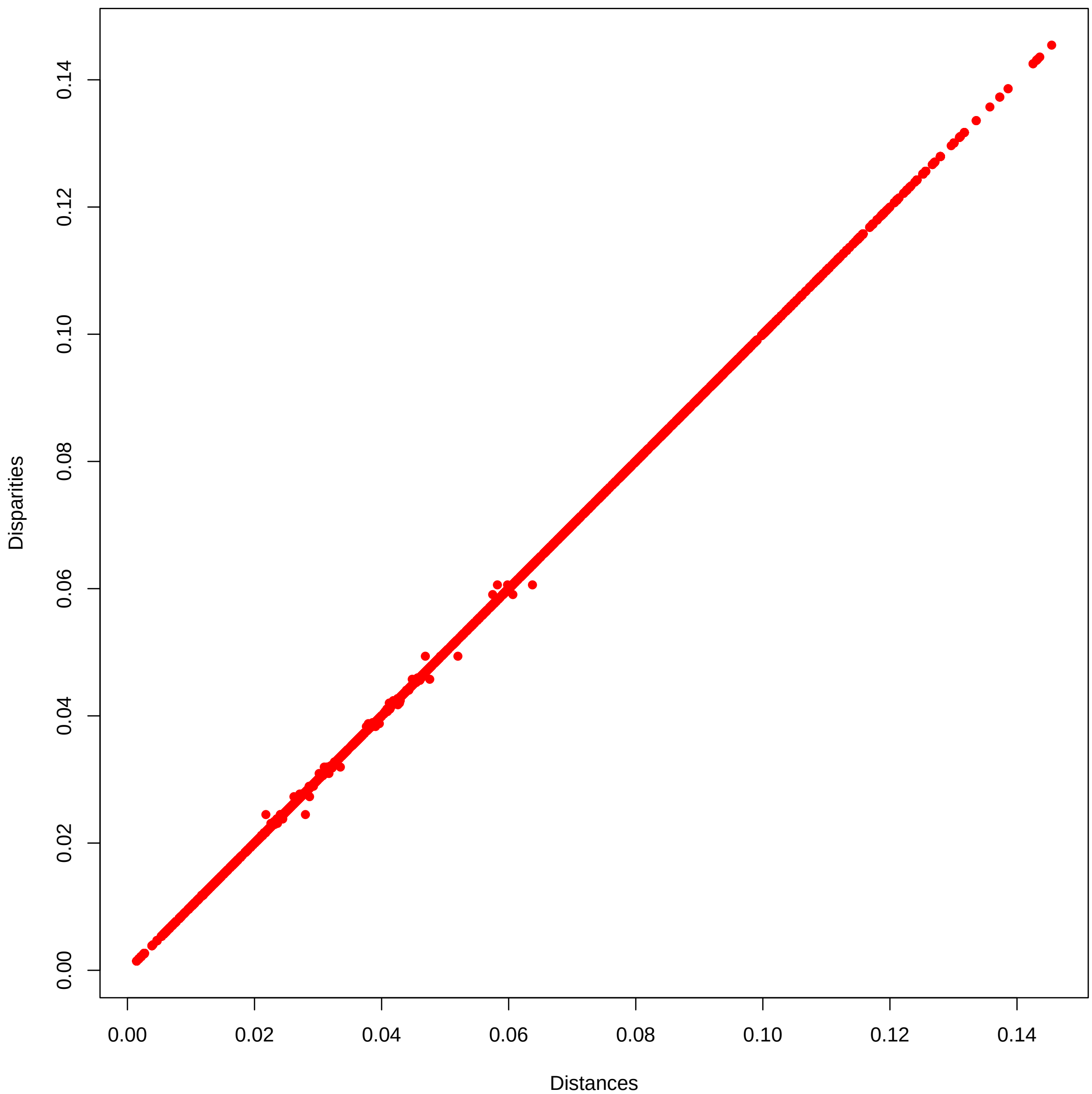


Figure 3: Galo D-Dhat Plot VHA

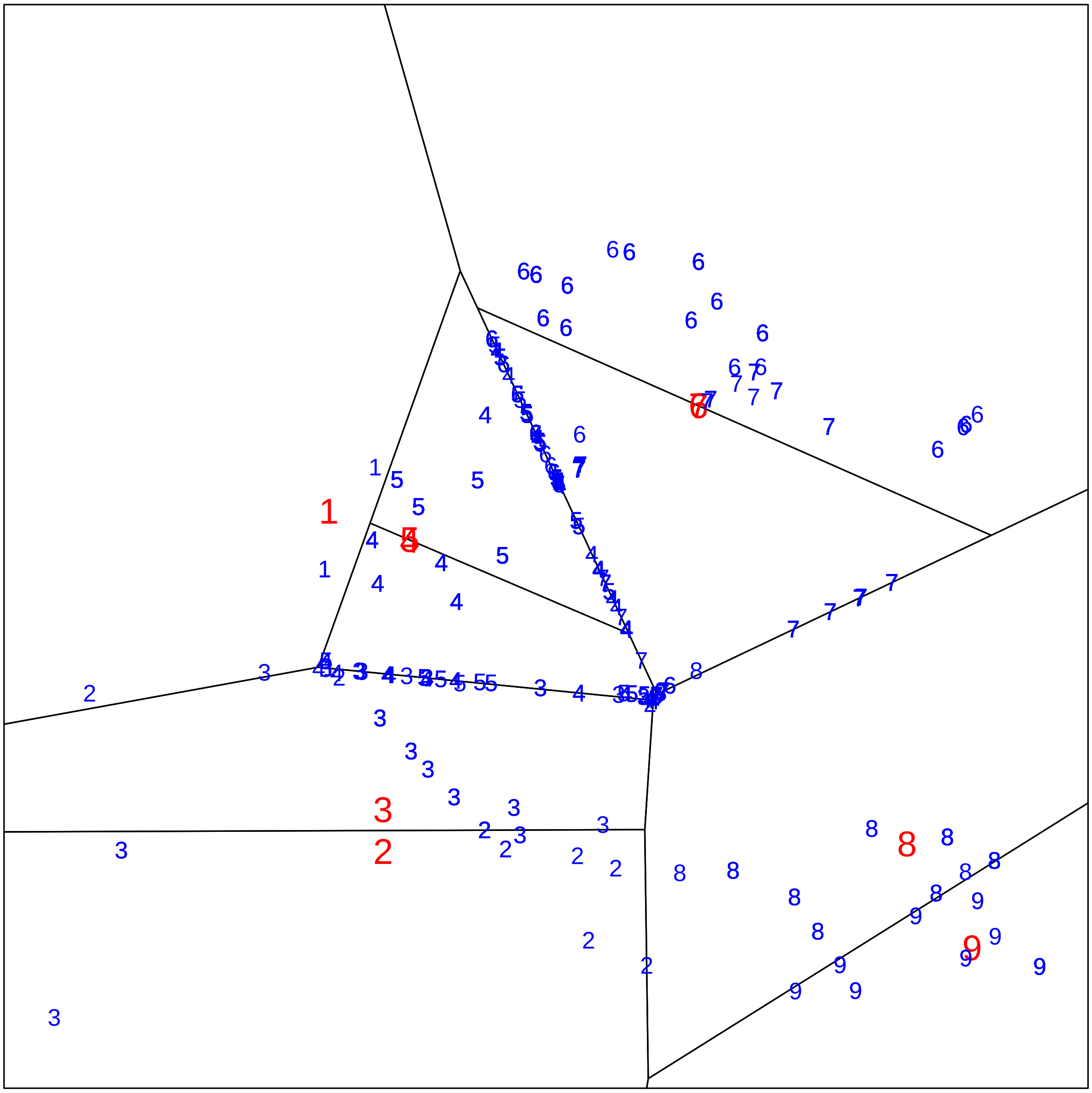


Figure 4: Galo IQ VHA

Figure 4 shows some characteristic properties of Voronoi plots. If the objects in two categories are not well separated the object-points will be on the perpendicular bisector of the category pair. This is partial degeneracy due to requiring weak monotonicity. The algorithm cannot impose strict monotonicity by placing an object in the interior of its Voronoi region, and compromises by putting it on the boundary. Thus the VHA plot will show object-points on lines. Some category scores are very close, which makes the Voronoi diagram unstable. The solution for the category-scores for IQ is

```
         dim1          dim2 freq
1 -0.03551673  0.0038947751    5
2 -0.02703754 -0.0491982427   17
3 -0.02706027 -0.0427253923  121
4 -0.02290451 -0.0006391014  234
5 -0.02289473 -0.0006162192  390
6  0.02216000  0.0203127031  244
7  0.02216072  0.0203143475  197
8  0.05467148 -0.0480676414   63
9  0.06483907 -0.0642872615   19
```

We see that categories 2 and 3 map into basically the same point. The same is true for categories 4 and 5, and for categories 6 and 7. So instead of nine points there are really only six. We can clean up the Voronoi diagram by using only those six categories. Note that we do not recode the data and redo the analysis, we just use the same plot with the same object-scores and just change the diagram. The result is in Figure 5.

5.3 Cetacea

The data in Vescia (1985) describe 36 whales, porpoises and dolphins using 15 categorical variables. In the github voronoi repository the data are in data/cetacea.R. Appendix C gives the common names of the 36 species and the names of the variables.

The variables are mostly physical characteristics, but they also include habitat and feeding. The average number of categories per variable is about four, and almost all variables are "nominal" (i.e. not "ordinal" and not "numerical"). In the same book Van der Burg (1985) presents a secondary analysis of the data with HGA and Guttman (1985) applies his MSA-I technique (which uses HGA as an initial configuration for its iterations).

We start with an HGA of the cetacea data. The initial value of the loss function … is 0.9945577132.

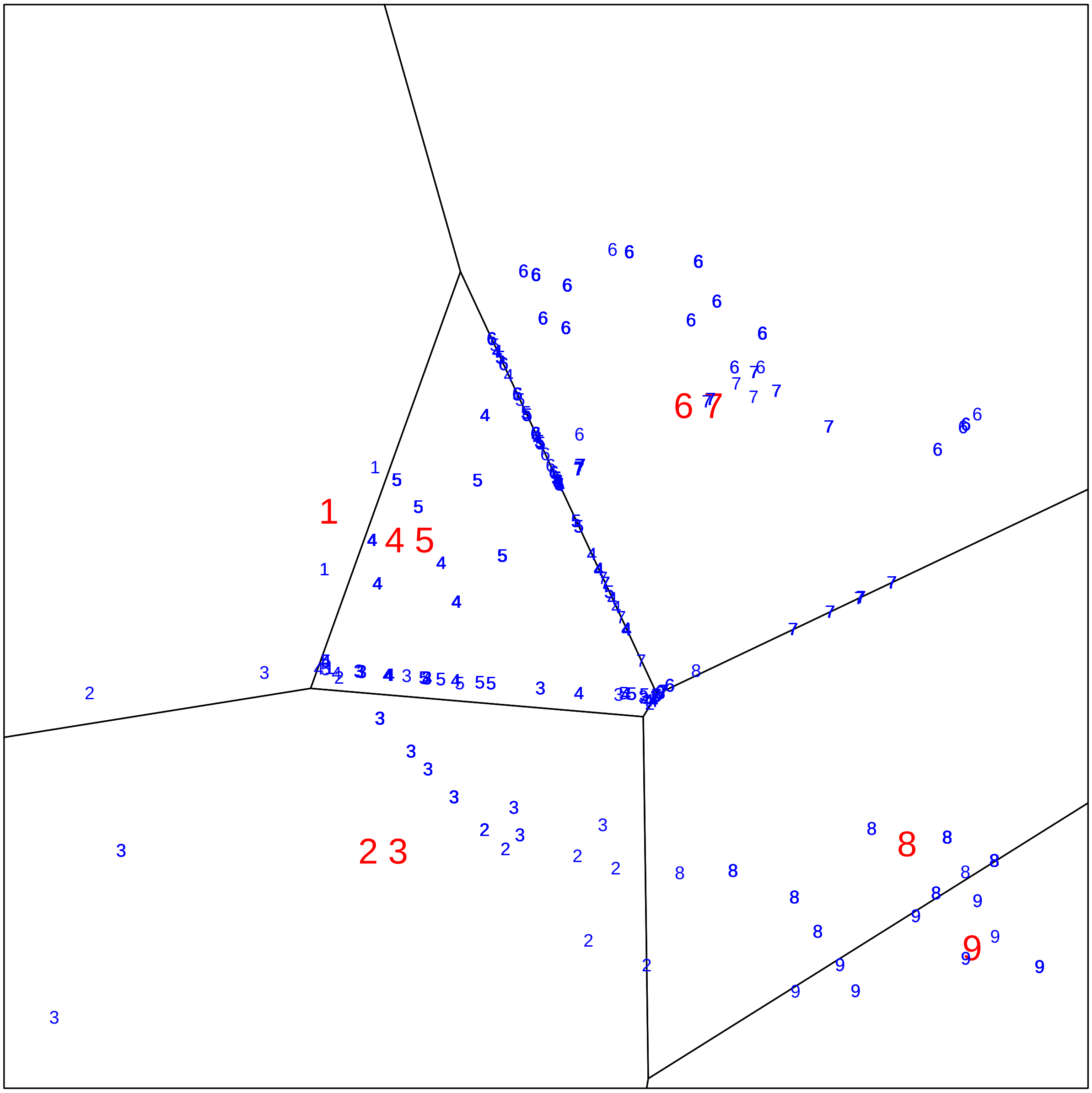


Figure 5: Galo IQ Corrected VHA

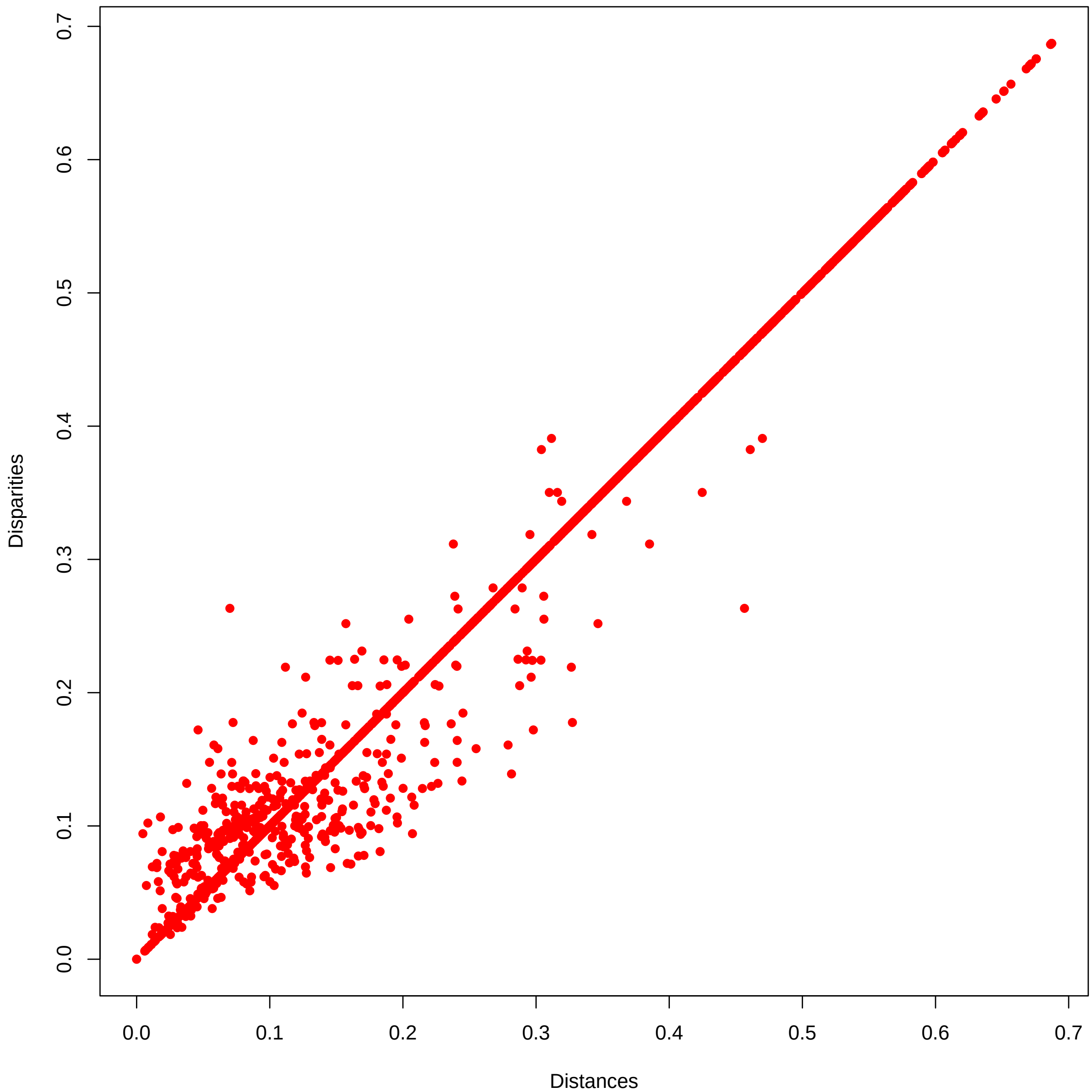


Figure 6: Cetacea D-Dhat Plot HGA

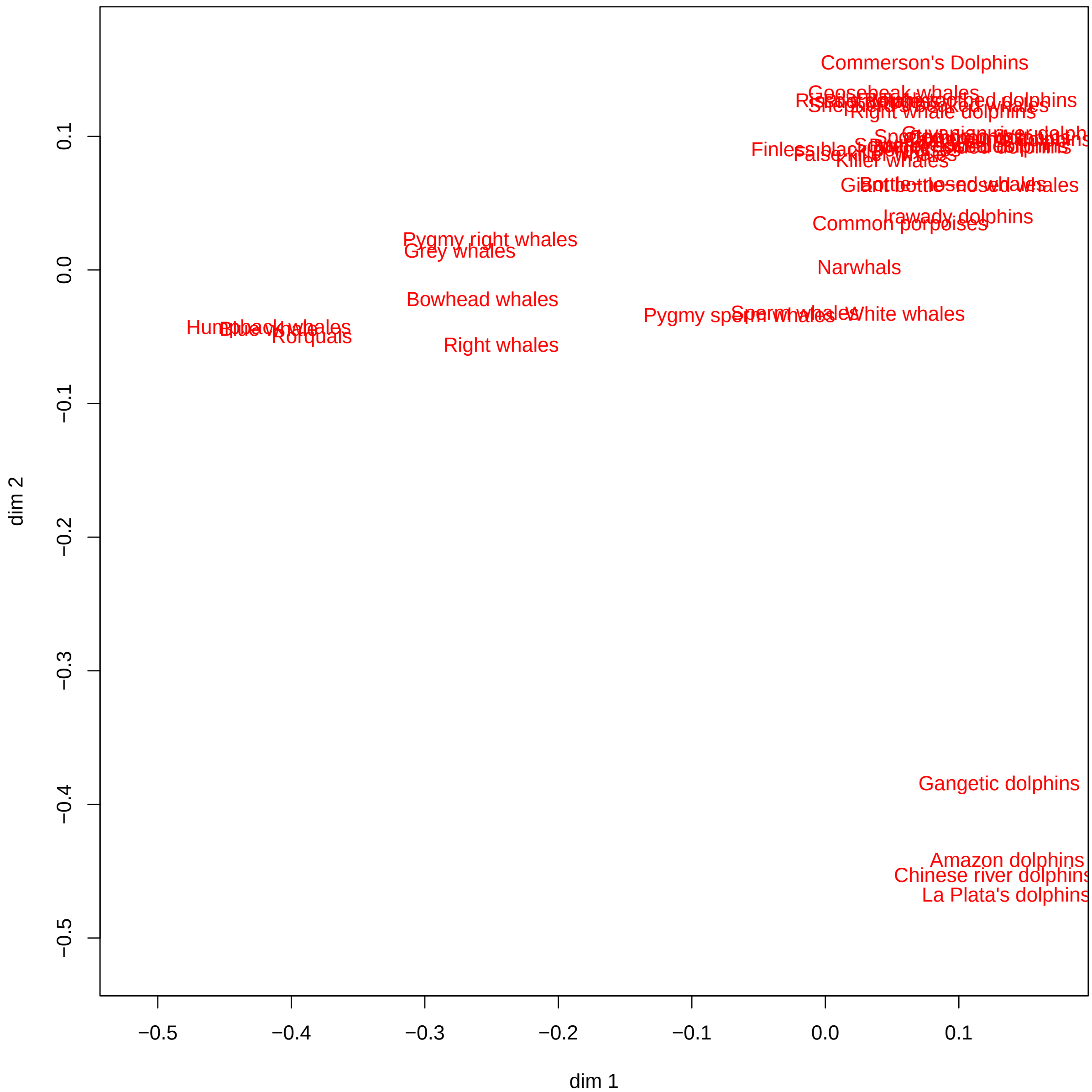


Figure 7: Cetacea Object Scores HGA

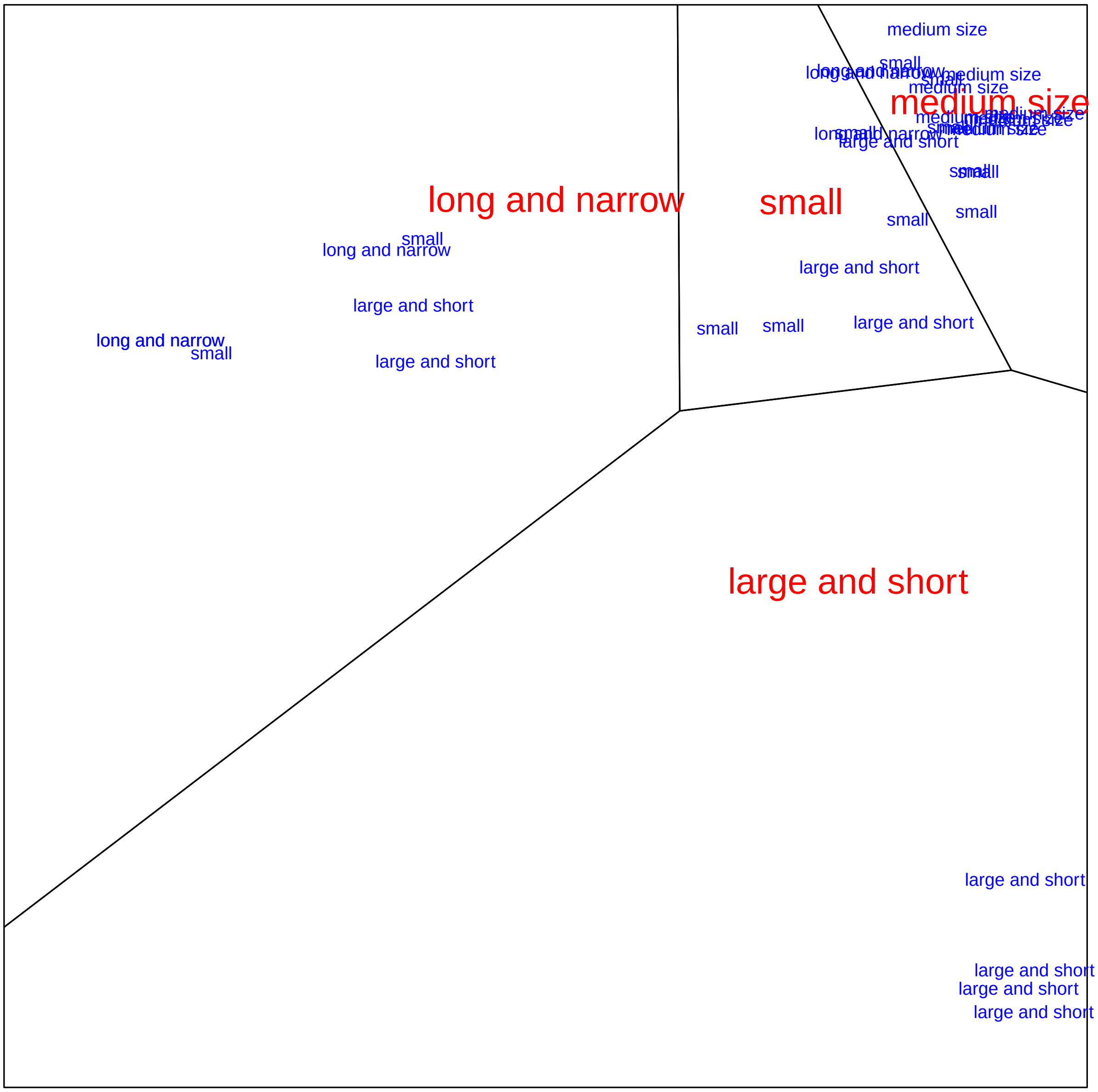


Figure 8: Cetacea Flippers HGA

Loss is 0.0000964845 after 1000 iterations.

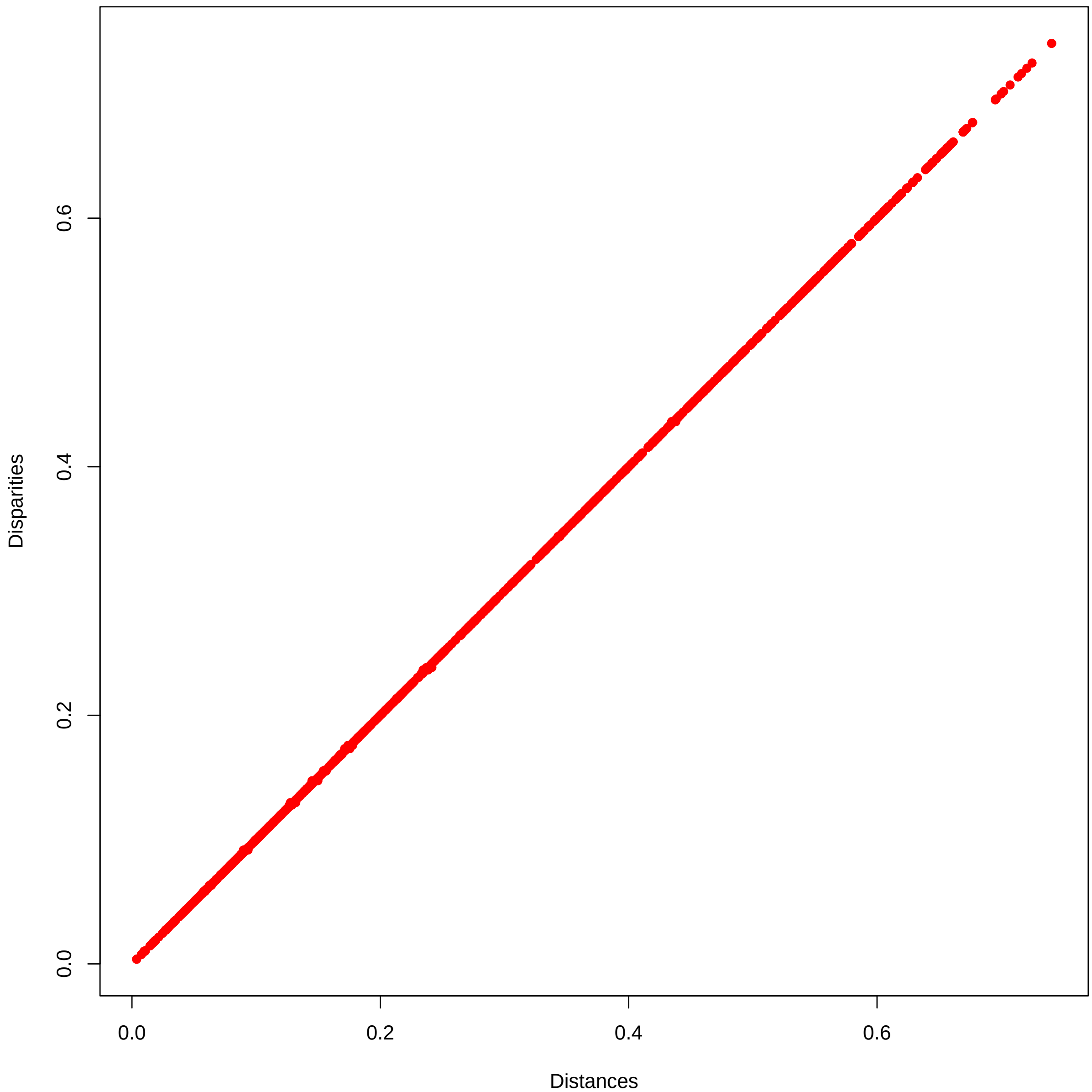


Figure 9: Cetacea D-Dhat Plot VHA

	dim1	dim2	freq
small	-0.03818544	-0.01751674	12
large and short	-0.03733595	-0.03078537	9
medium size	0.18051322	0.15150857	9
long and narrow	-0.08505453	0.10776408	6

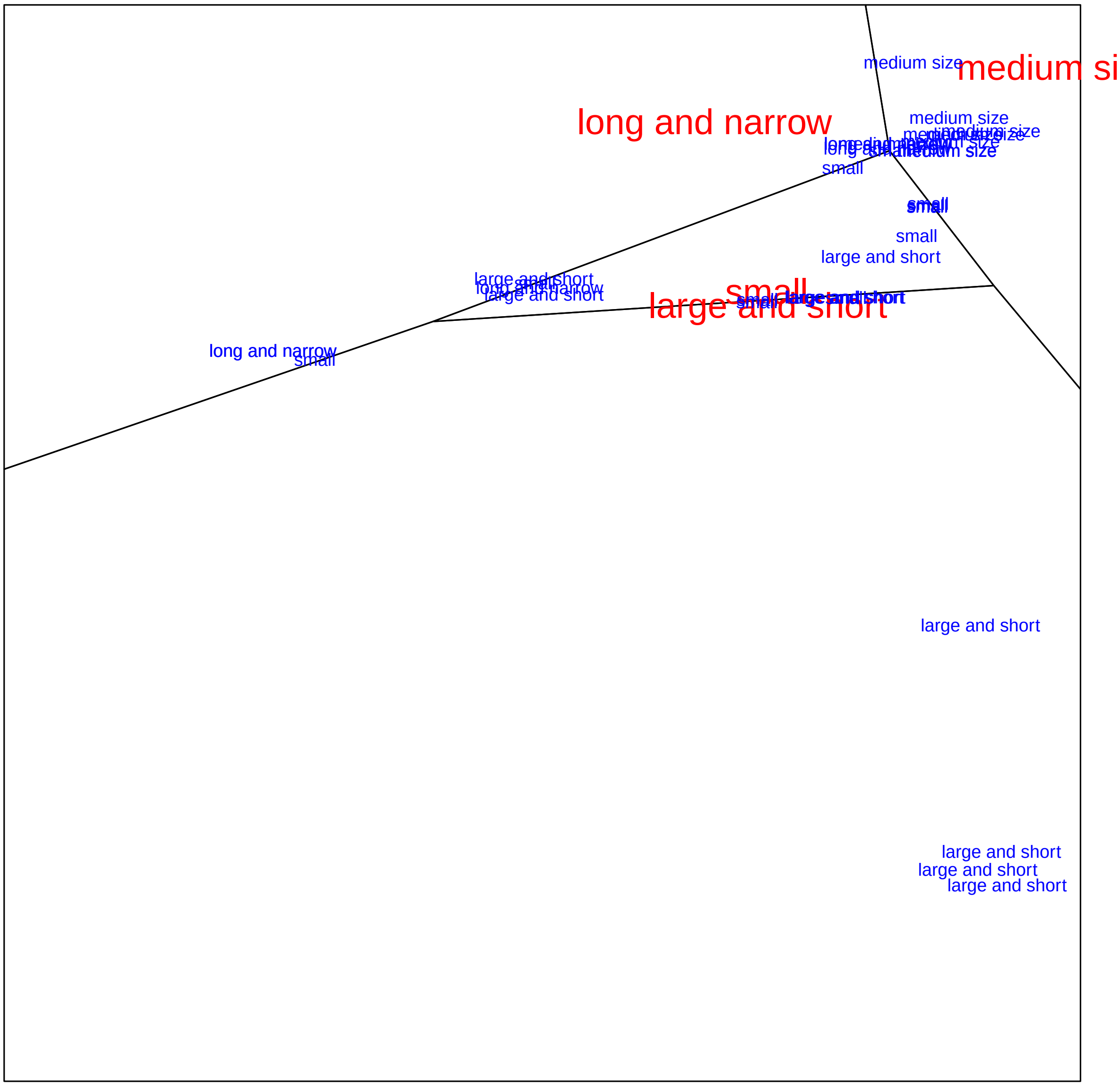


Figure 10: Cetacea Flippers

Loss is 0.0003024926 after 1000 iterations.

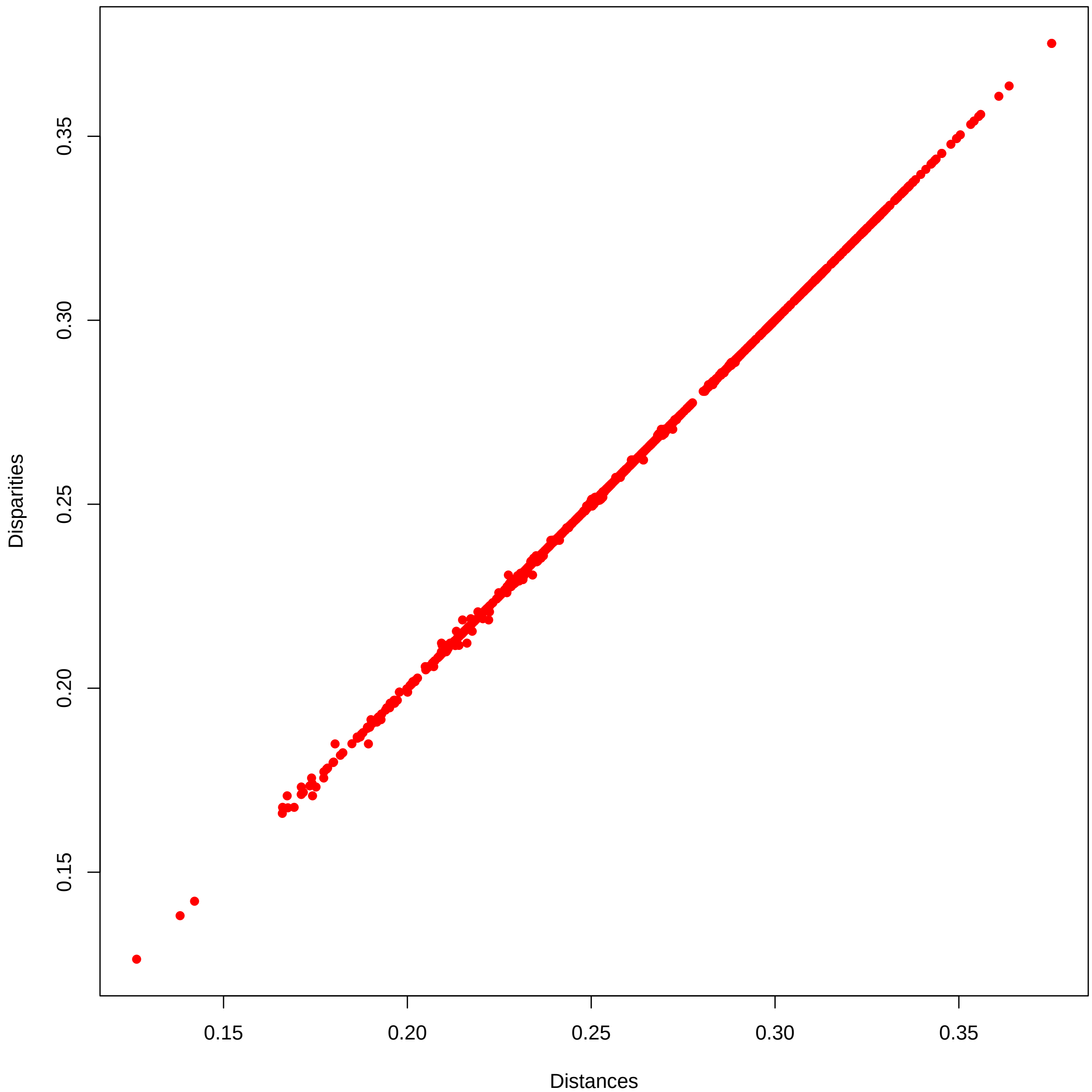


Figure 11: Cetacea D-Dhat Plot VHA XX

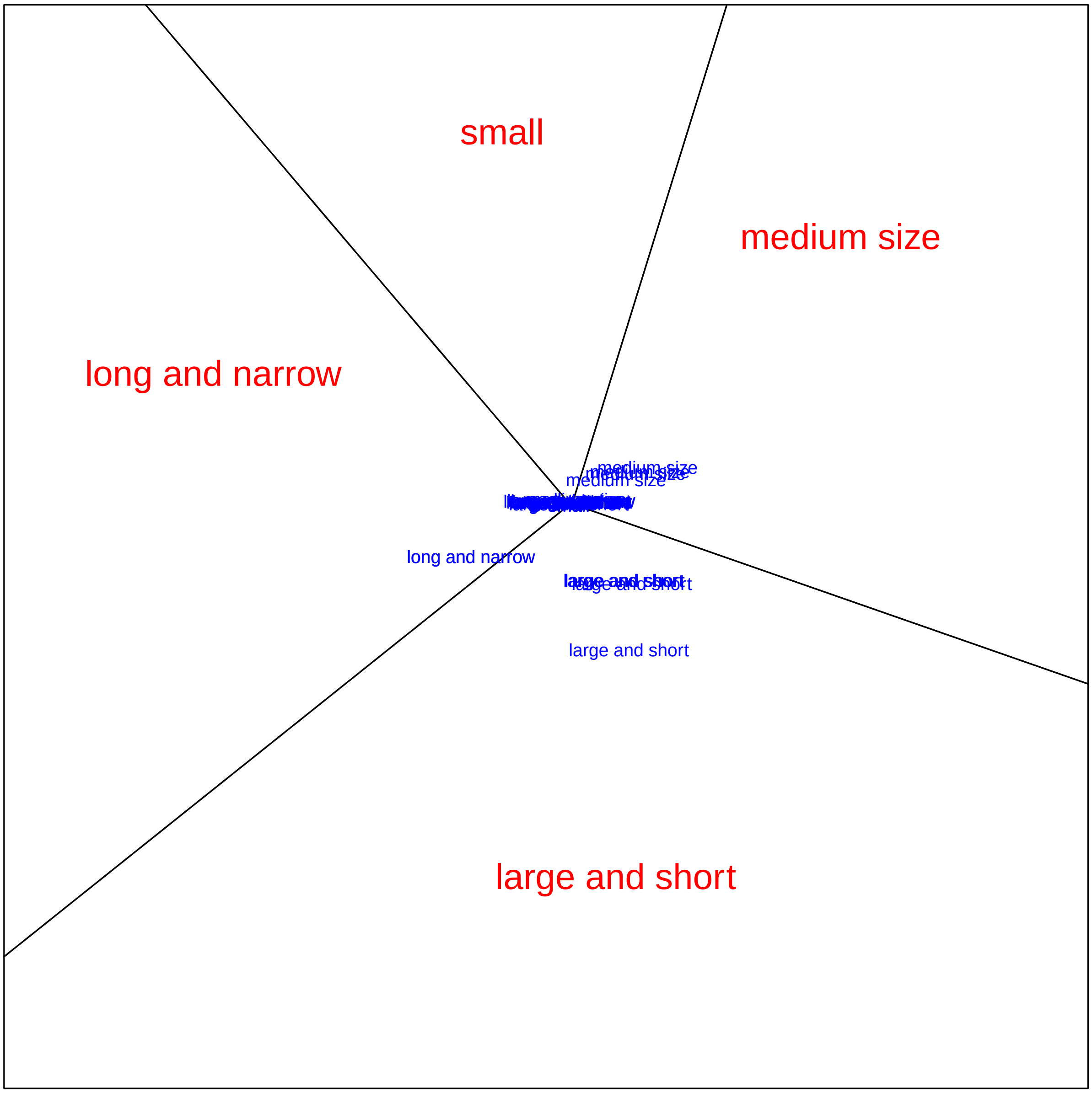


Figure 12: Cetacea Flippers VHA XX

6 Appendix: Cetacea

```
 [1]           Bowhead whales                 Rorquals
 [3]               Blue whale Giant bottle-nosed whales
 [5]     Commerson's Dolphins             White whales
 [7]          Common dolphins              Grey whales
 [9]             Right whales             Pilot whales
[11]          Risso's dolphins     Bottle-nosed whales
[13]          Amazon dolphins        Pygmy sperm whales
[15]     White-sided dolphins    Chinese river dolphins
[17]     Right whale dolphins           Humpback whales
[19]          Sowerby's whales                 Narwhals
[21]        Pygmy right whales  Finless black porpoises
[23]          Irawady dolphins            Killer whales
[25]          Common porpoises             Sperm whales
[27]         Gangetic dolphins       False killer whales
[29]   Guyanian river dolphins       Cameroun's dolphins
[31]          Spotted dolphins    Rough toothed dolphins
[33]       La Plata's dolphins  Shepherd's beaked whales
[35]     Bottle-nosed dolphins          Goosebeak whales
```

```
 [1]                               NECK                    FORM OF THE HEAD
 [3]                  SIZE OF THE HEAD                                 BEAK
 [5]                        DORSAL FIN                             FLIPPERS
 [7]                      SET OF TEETH                              FEEDING
 [9]                         BLOW HOLE                                COLOR
[11]                CERVICAL VERTEBRAE           LACRYMAL AND JUGAL BONES
[13]                           HABITAT LONGITUDINAL FURROWS ON THE THROAT
[15]                        HEAD BONES
```

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